CHETANA MURALIDHARAN

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Education

San Jose State University San Jose, CA
M.S., Applied Data Intelligence GPA: 3.88/4.00 *August 2024 – May 2026*

Visvesvaraya Technological University KA, India
B.E., Computer Science & Engineering CGPA: 8.5/10.00 *August 2017 – August 2021*

Experience

Instructional Student Assistant(ISA) - DATA 266: Generative Models January 2026 – May 2026
San Jose State University *San Jose, CA*

- Designed 14 assignments and 2 labs spanning Transformer architecture, CUDA programming, diffusion models, LoRA/PEFT fine-tuning, RAG, and multimodal agentic systems, and delivered in-class technical demos to 35+ graduate students.
- Debugged 100+ student PyTorch pipelines and multimodal AI workflows spanning Hugging Face model integration, vector database indexing, and LangChain agent design, improving end-to-end implementation reproducibility across the cohort.

Data Engineer (ETL Developer) August 2021 – July 2024
Accenture *India*

- Cut pipeline runtime by 85% (6-8 hrs to 45 min) by re-architecting 60+ full-refresh loads into watermark-based CDC incremental pipelines on Snowflake and AWS S3, enabling KPI reporting across 7 APAC markets with idempotent reruns, zero data gaps.
- Reduced new distributed source onboarding from days to minutes by designing a metadata-driven, config-table-controlled pipeline architecture that integrated 6+ heterogeneous systems (SQL Server, Oracle, SAP HANA, S3, SFTP) into a unified Snowflake warehouse without code changes per new source.
- Eliminated manual data validation for 60+ objects per pipeline run by building an automated post-load reconciliation framework with configurable per-object DQ metrics (row count, distinct key count, sum of amounts), ensuring source-to-target accuracy.
- Increased data accuracy across all pipelines by implementing soft-delete detection via primary key anti-joins, MD5-based composite key deduplication, and automated source-to-target reconciliation, establishing DQ standards for downstream BI consumption.

Projects

LineageIQ: dbt Observability and AI Blast Radius Platform April 2026 – June 2026

- Enforced data quality across 9 dbt models by implementing 15+ automated tests covering uniqueness, referential integrity, null checks, and accepted values, preventing bad data from propagating to downstream dashboards and consumers.
- Cut blast radius triage from minutes to seconds by feeding Airflow-orchestrated schema drift detections and lineage depth metrics into an LLM, producing a CRITICAL-to-LOW ranked impact report per schema change.
- Automated schema drift detection across a Snowflake schema registry by baseline-comparing column definitions after every Airflow-orchestrated dbt build, flagging HIGH-severity column drops and type changes before pipeline failures could occur.

WeaveNet: Communication Analytics Platform August 2025 – May 2026

- Reduced dataset noise by 56% and retained 226K high-quality records from 517K+ enterprise emails by building modular parsing, deduplication, topic enrichment, and thread reconstruction pipelines, producing a clean analytical foundation for graph modeling.
- Designed scalable data infrastructure supporting 764K+ nodes and 2.86M relationships by modeling data into structured node and edge schemas in Python with indexed retrieval layers, enabling reusable analytical layers for large-scale downstream consumers.
- Reduced graph query latency to under 100ms across 40+ traversal and analytical queries by implementing indexed data structures and optimized retrieval workflows, improving time-to-insight for downstream consumers at scale.

Smoke & Smog: Fire Incidents & Air Quality Analytics Pipeline February 2025 – May 2025

- Built an end-to-end ELT platform processing 4M+ environmental records from NASA, weather, and AQ sources using Airflow, dbt, and Snowflake, producing validated OLAP-ready data mart models across 25 U.S. cities.
- Strengthened pipeline reliability across 7 dbt transformation models by implementing 15+ data quality tests and freshness checks, enforcing a semantic layer of validated, reusable analytical assets aligned with DataOps and CI/CD best practices.
- Delivered two Power BI dashboards with DAX measures and a composite Environmental Distress Score derived from 6 lag-based AQI metrics, enabling faster identification of high-risk regions for business stakeholders.

Technical Skills

Data Engineering: ETL/ELT Pipelines, CDC Pipelines, dbt (Models, Tests, Snapshots, Freshness Checks), Incremental Processing, Real-Time Pipelines, Apache Airflow, Apache Spark, Kafka, Orchestration Frameworks, Automation Workflows, Pipeline Design
Analytics Engineering: Dimensional Modeling (Star/Snowflake Schema), Semantic Layer Design, Data Quality Testing, Schema Drift Detection, Pipeline Observability, DataOps, Self-Service Analytics, KPI Development
Data Warehousing and Cloud: Snowflake (SnowPro Core Certified), AWS S3, AWS Redshift, BigQuery, Databricks, Data Lakes, MySQL, PostgreSQL, Fivetran
BI & Visualization: Power BI, DAX, Tableau, Streamlit, Plotly, Dashboard Development, KPI Reporting
Compliance and Access Controls: PHI Data Handling, HIPAA-Aligned Access Controls, Data Security Best Practices, Data Integrity, Schema Enforcement, Data Reconciliation, Privacy and Regulatory Compliance
AI & LLM Tools: Prompt Engineering, RAG, LangChain, Agentic Workflows, Vector Databases, GraphRAG, NetworkX
Languages & Dev Tools: Python, SQL, Java, Git, CI/CD, Agile/Scrum, Cross-Functional Collaboration